WEEK 3 REPORT
STOCK MARKET SALES REPORT DASHBOARD

Power BI Project

1. Introduction

This report documents the development of a Stock Market Sales Report Dashboard in Microsoft Power BI. The project covers dataset import and preparation, creation of a Date Table, calculated date columns, data relationships, DAX measures, time-intelligence calculations, and dashboard design.

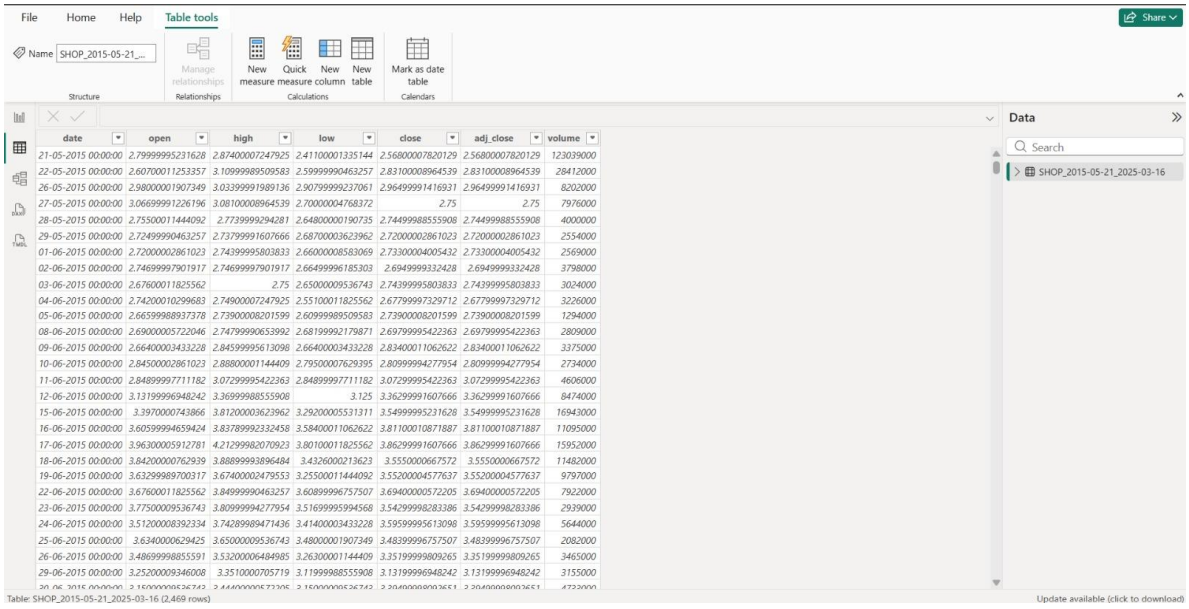
The dashboard presents historical stock information in a visual format. It helps users review stock prices and trading activity, compare values over time, and identify patterns in the available data.

2. Dataset Import and Preparation

The Shopify stock market dataset was imported into Power BI Desktop for analysis and visualization. It contains historical stock information from May 2015 to March 2025, including opening price, closing price, highest price, lowest price, adjusted closing price, and trading volume.

Power Query Editor was used to inspect the dataset structure and validate records. Data types were checked so that dates, prices, and volume values could be handled correctly. The date column was duplicated and converted to a Date format to support a relationship with the Date Table.

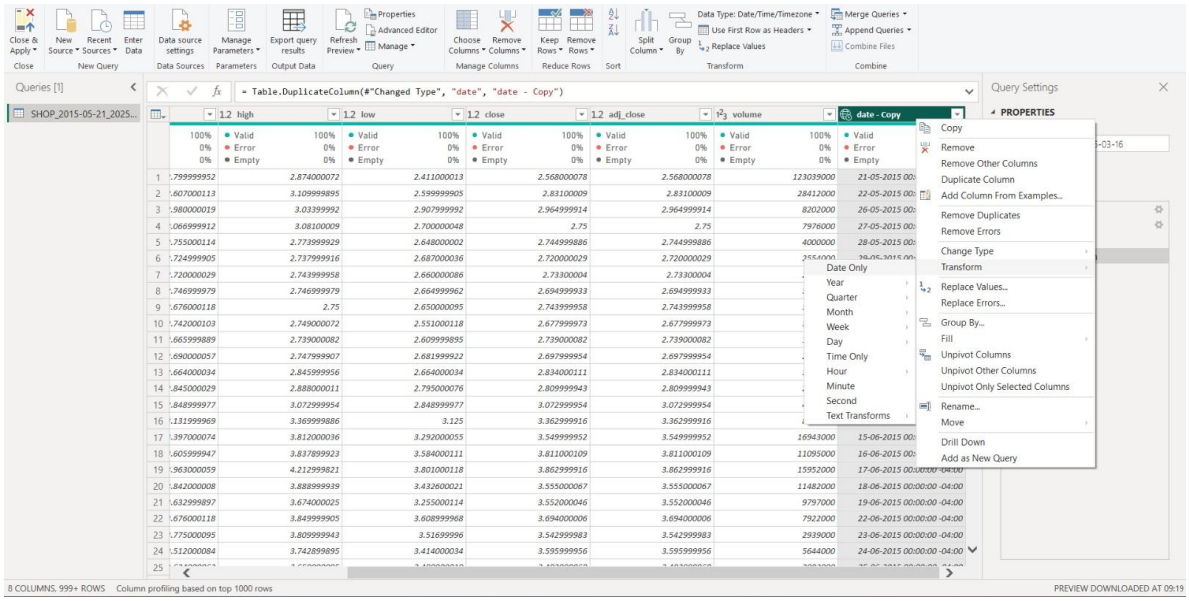
Activities included importing the dataset, checking column data types, reviewing data quality and consistency, preparing the date field, and loading the cleaned data into the Power BI model.



date	open	high	low	close	adj_close	volume
21-05-2015 00:00:00	2.799999952316280	2.874000072479250	2.411000013351440	2.568000078201290	2.568000078201290	123039000
22-05-2015 00:00:00	2.607000112533570	3.109999895095830	2.599999904632570	2.831000089645390	2.831000089645390	28412000
26-05-2015 00:00:00	2.980000019073490	3.033999919891360	2.907999992370610	2.964999914169310	2.964999914169310	8202000
27-05-2015 00:00:00	3.066999912261960	3.081000089645390	2.700000047683720	2.75	2.75	7976000
28-05-2015 00:00:00	2.755000114440920	2.773999929428100	2.648000001907350	2.744999885555080	2.744999885555080	4000000
29-05-2015 00:00:00	2.724999904632570	2.737999916076660	2.687000036239620	2.720000028610230	2.720000028610230	2554000
01-06-2015 00:00:00	2.720000028610230	2.743999958038330	2.660000085830690	2.733000040054320	2.733000040054320	2569000
02-06-2015 00:00:00	2.746999979019170	2.746999979019170	2.664999961853030	2.694999933242280	2.694999933242280	3798000
03-06-2015 00:00:00	2.676000118255620	2.75	2.650000095367430	2.743999958038330	2.743999958038330	3024000
04-06-2015 00:00:00	2.742000102996830	2.749000072479250	2.551000118255620	2.677999973297120	2.677999973297120	3226000
05-06-2015 00:00:00	2.665599889373780	2.739000082015990	2.609999895095830	2.739000082015990	2.739000082015990	1294000
08-06-2015 00:00:00	2.690000057220460	2.747999906539920	2.681999921798710	2.697999954224630	2.697999954224630	2809000
09-06-2015 00:00:00	2.664000034332280	2.845999956130980	2.664000034332280	2.834000110626220	2.834000110626220	3375000
10-06-2015 00:00:00	2.845000028610230	2.888000011444090	2.795000076293950	2.809999942779540	2.809999942779540	2734000
11-06-2015 00:00:00	2.848999977111820	3.072999954223630	2.848999977111820	3.072999954223630	3.072999954223630	4606000
12-06-2015 00:00:00	3.131999969482420	3.369999885559080	3.125	3.362999916076660	3.362999916076660	8474000
15-06-2015 00:00:00	3.39700000743866	3.812000036239620	3.292000055313110	3.549999952316280	3.549999952316280	16943000
16-06-2015 00:00:00	3.605999946594240	3.837899923324580	3.584000110626220	3.811000108718870	3.811000108718870	11095000
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19-06-2015 00:00:00	3.632999897003170	3.674000024795530	3.255000114440920	3.552000045776370	3.552000045776370	9797000
22-06-2015 00:00:00	3.676000118255620	3.849999904632570	3.608999967575070	3.694000005722050	3.694000005722050	7922000
23-06-2015 00:00:00	3.775000095367430	3.809999942779540	3.516999959945680	3.542999982833860	3.542999982833860	2939000
24-06-2015 00:00:00	3.512000083923340	3.742899894714360	3.414000034332280	3.595999956130980	3.595999956130980	5644000
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26-06-2015 00:00:00	3.486999988555910	3.532000064849850	3.263000011444090	3.351999998092650	3.351999998092650	3465000
29-06-2015 00:00:00	3.252000093460080	3.35100007057190	3.119999885559080	3.131999969482420	3.131999969482420	3155000

2.1 Reviewing the Dataset

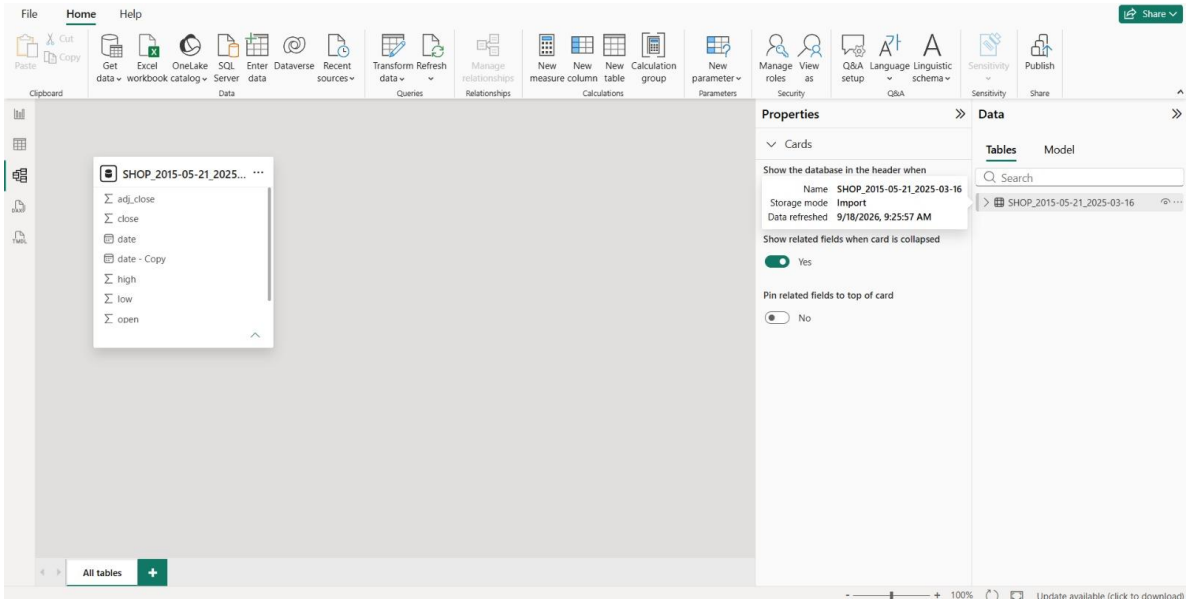
The dataset was reviewed in Power Query to understand its columns and structure before creating visuals. Correct data types are important because Power BI uses them to determine how fields can be aggregated, filtered, and calculated.



Dataset fields and preparation.

2.2 Checking Data Quality

Data quality checks help identify missing values, unexpected formats, and inconsistent records. Reviewing the data before loading helps ensure that the fields are suitable for analysis and reporting.

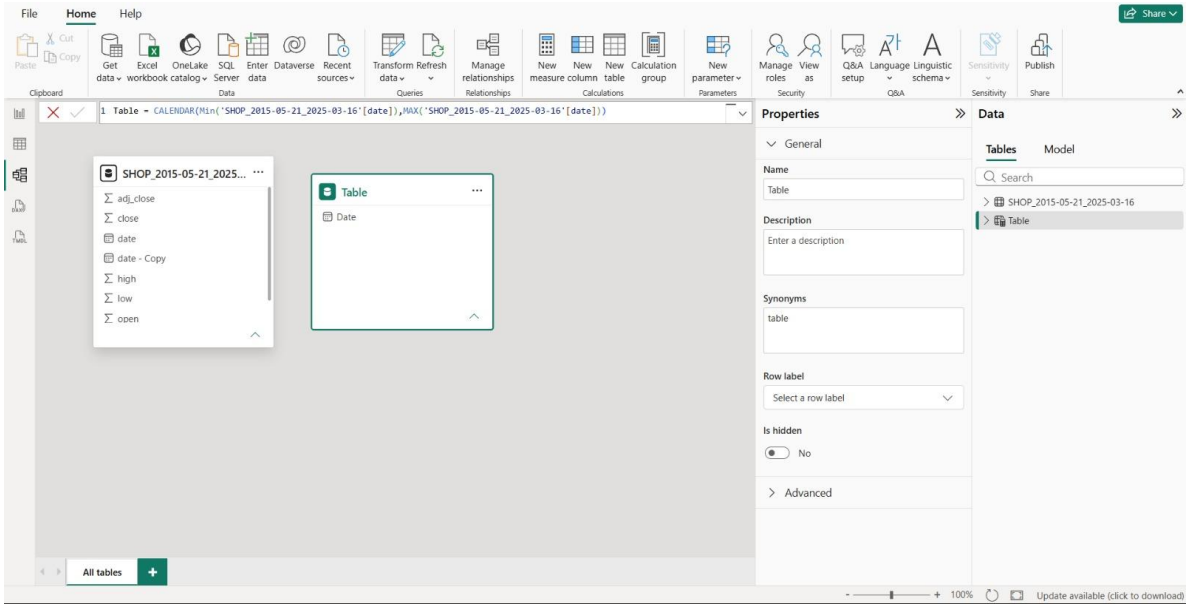


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3. Date Table Creation

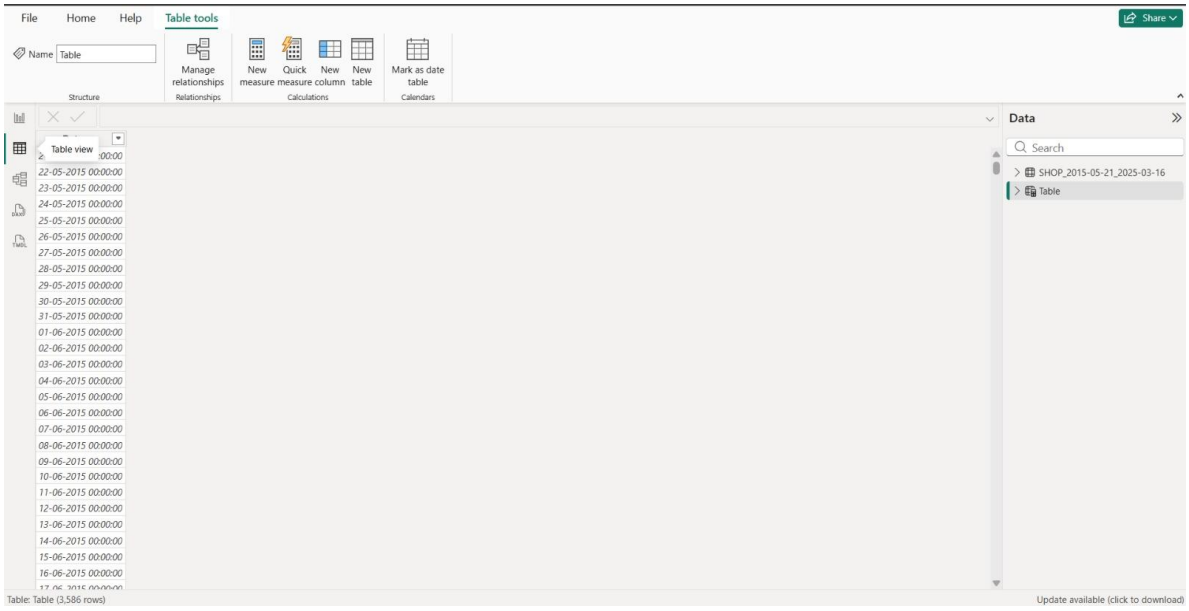
A dedicated Date Table was created to support time-based reporting and DAX time-intelligence functions. A continuous calendar table allows stock information to be filtered and summarized consistently by year, quarter, month, and day.

The Date Table was generated using the CALENDAR function, with the date range based on the minimum and maximum dates in the stock dataset. Additional calculated columns were created for Year, Quarter, Month, Month Number, and Day.



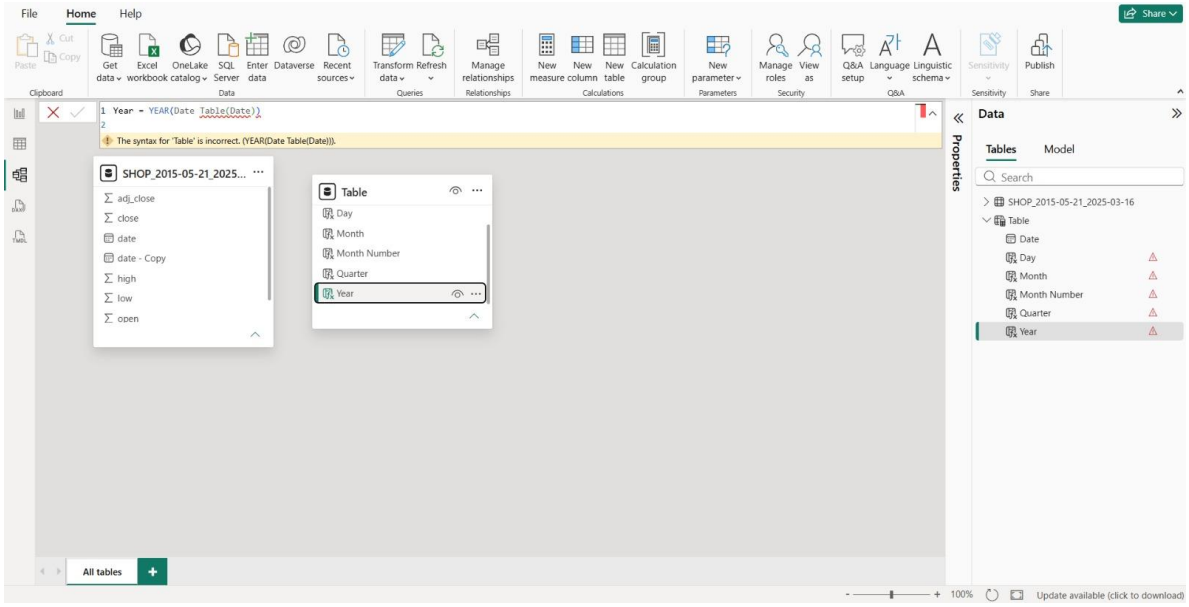
3.1 Date Table Structure

The Date Table provides a central calendar for the report. It organizes dates into useful reporting levels and supports consistent filtering across visuals. It should be connected to the stock dataset through matching date fields.



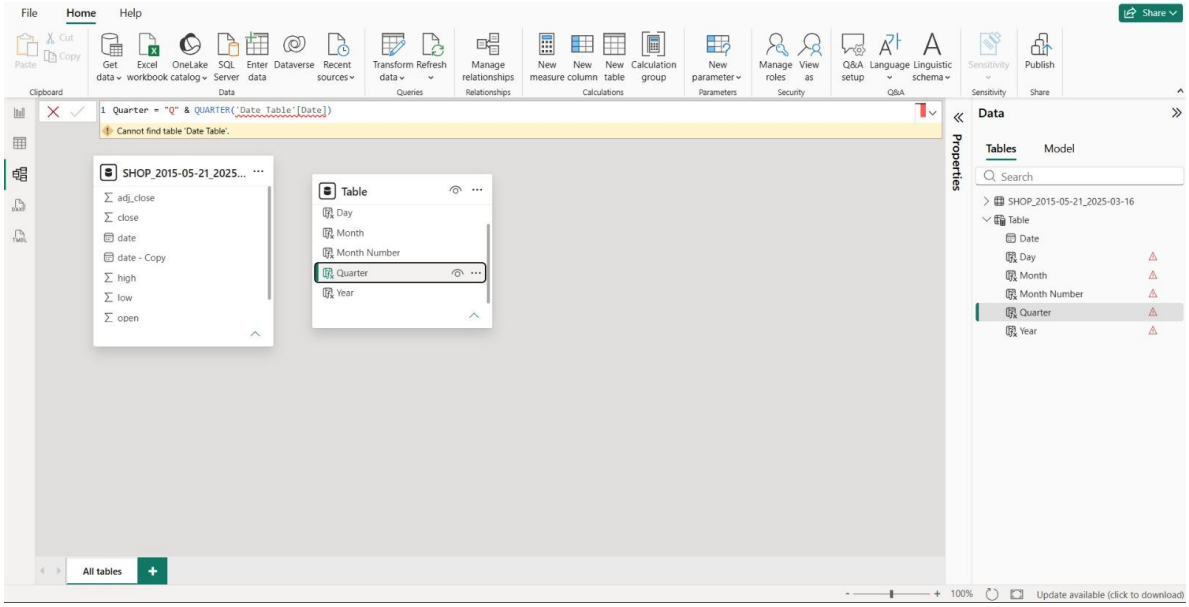
4. Calculated Date Columns

Calculated date columns make it possible to group and analyse stock data at different levels of detail. Year supports annual comparisons, Quarter supports quarterly reporting, Month supports monthly analysis, Month Number controls chronological month sorting, and Day provides daily-level detail.



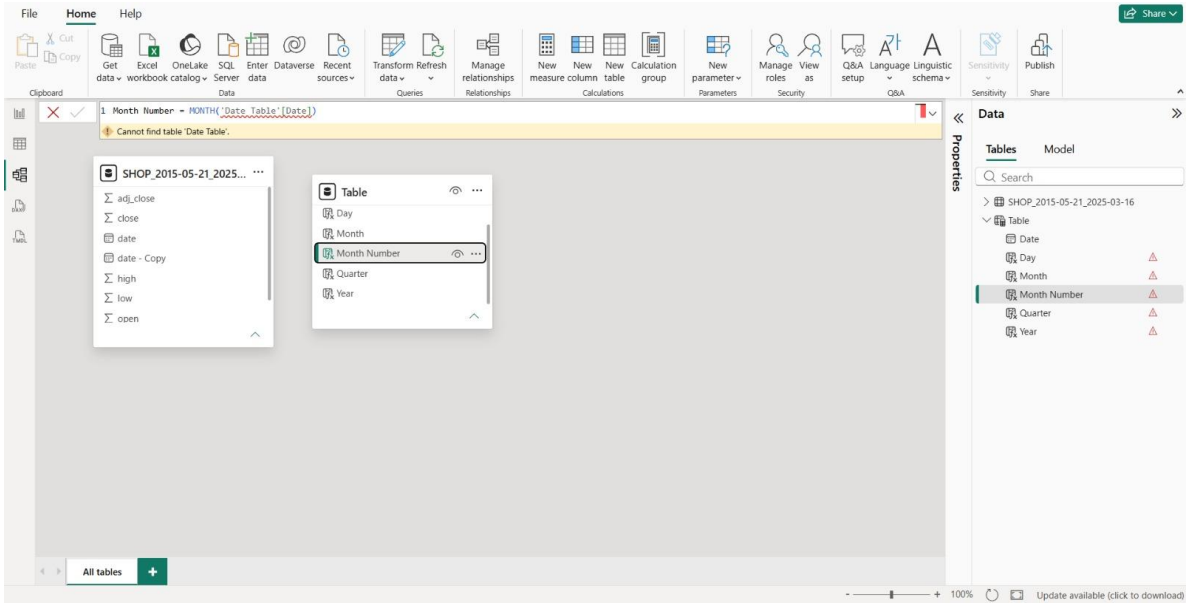
4.1 Quarter Column

The Quarter column categorizes dates into Q1, Q2, Q3, and Q4. Quarterly grouping allows users to compare stock performance across different parts of the year and use quarter-based filters in visuals.



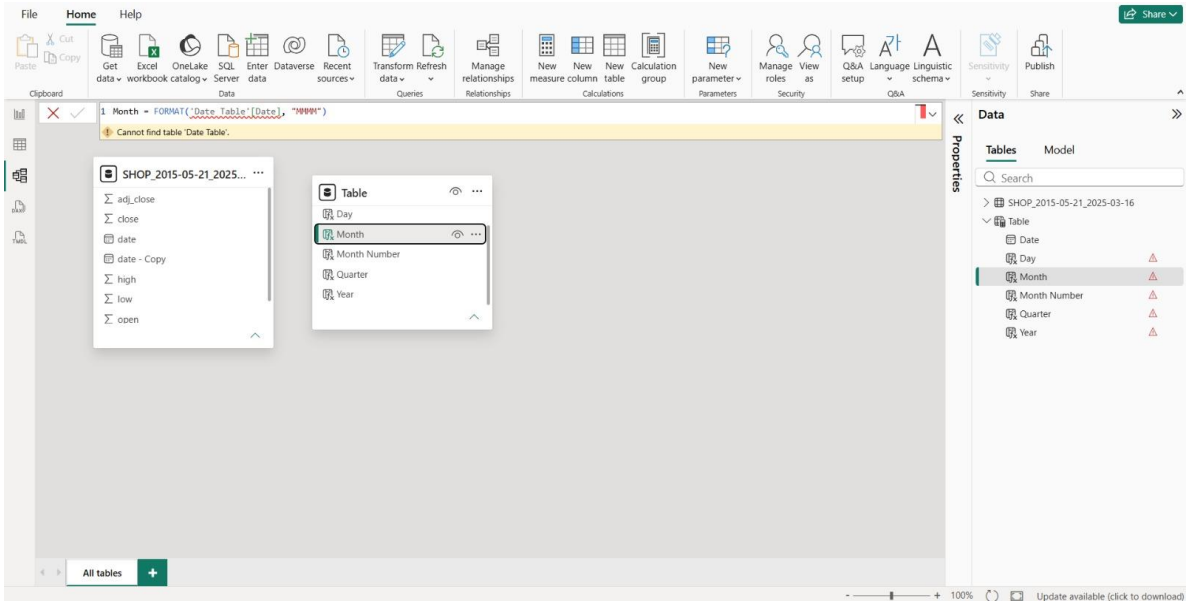
4.2 Month Number Column

The Month Number column stores each month as a number from 1 to 12. It is used to sort month names in calendar order rather than alphabetically, keeping trend charts and tables in chronological sequence.



4.3 Month Column

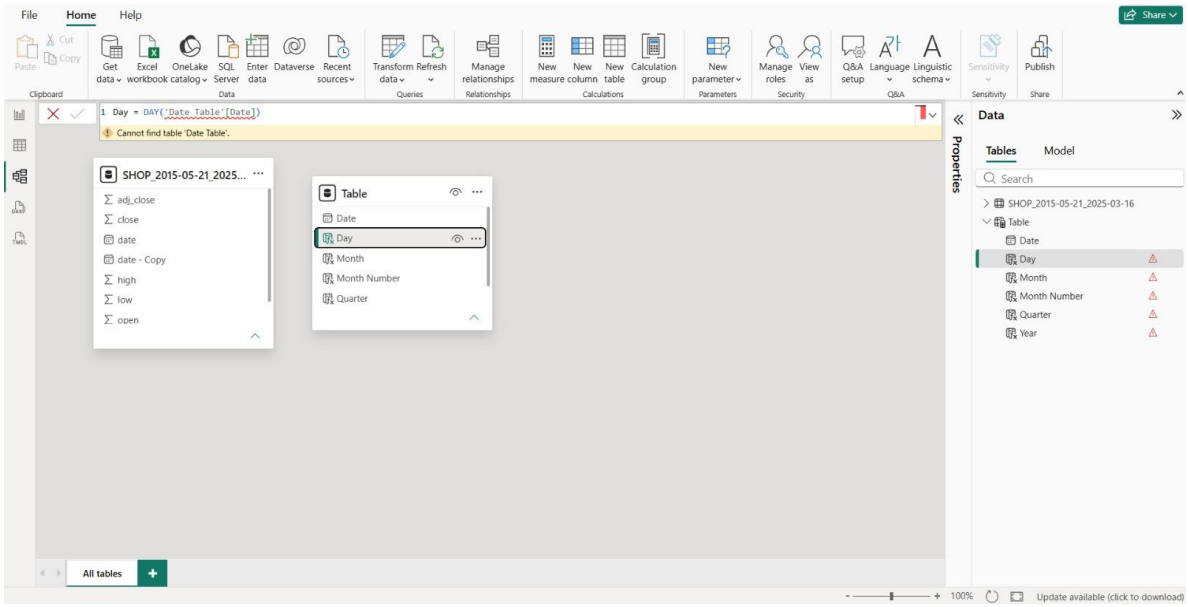
The Month column displays each month name in a readable format. When sorted by Month Number, visuals can show the months from January through December, making monthly stock trends easier to interpret.



5. Relationship and Model View

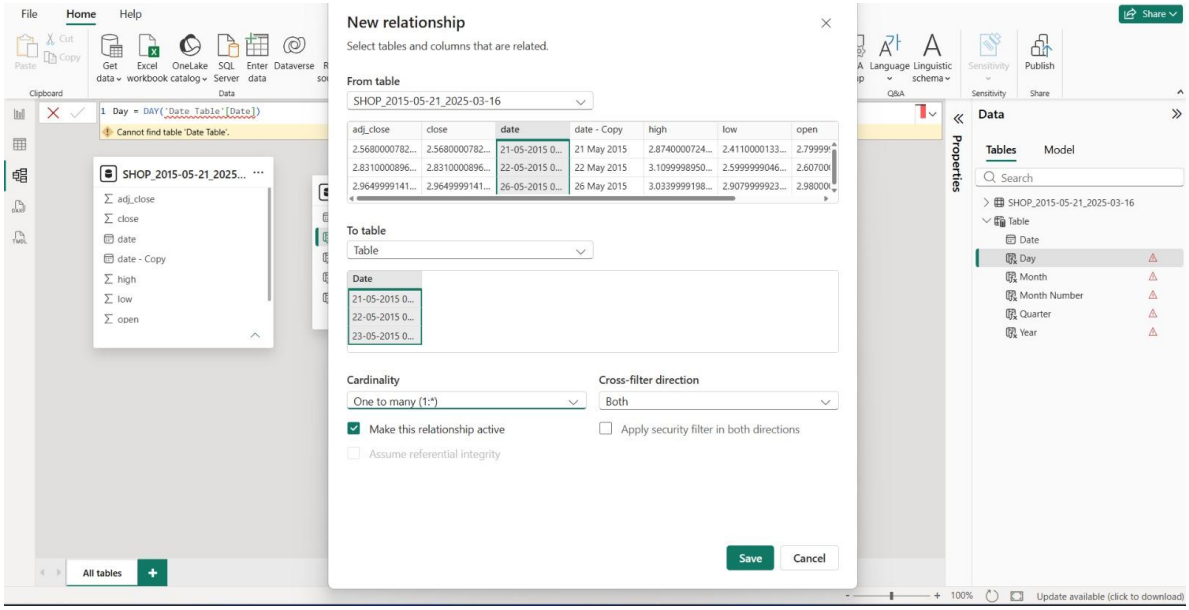
The model view is used to connect the stock dataset to the Date Table. A relationship between the date fields enables calendar filters to affect stock records and provides the foundation for time-based calculations.

A well-organized model reduces ambiguity and makes report calculations more reliable. The date fields used for the relationship must have compatible Date data types.



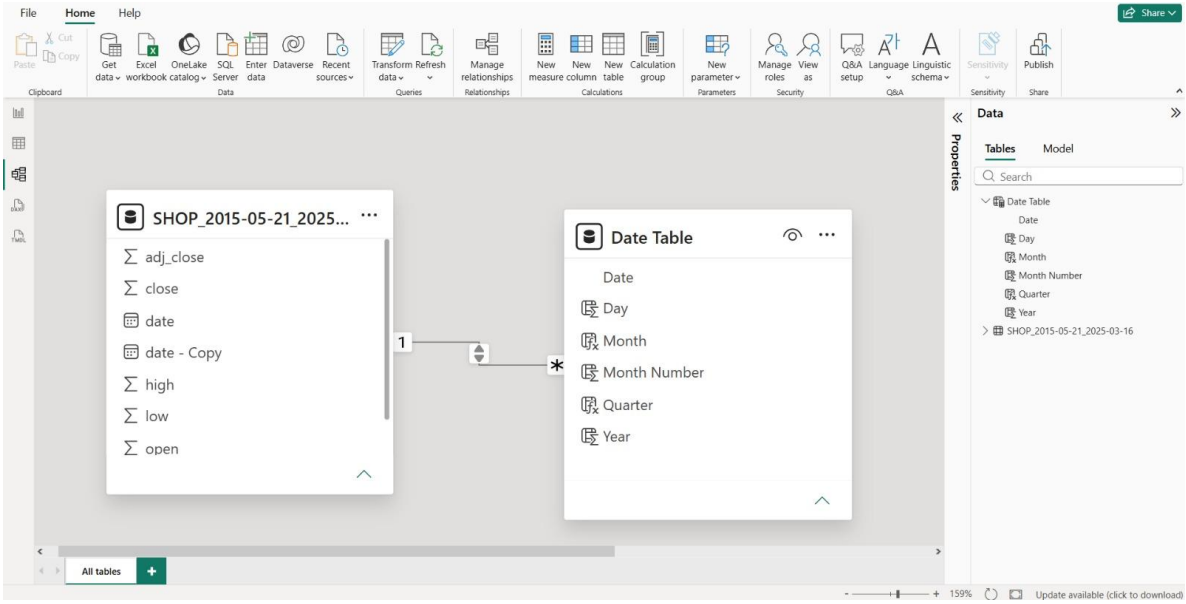
5.1 Reviewing the Data Model

The model view provides a visual representation of tables and their relationships. Checking this view helps confirm that the Date Table is connected to the stock data and that the intended filtering path is available.



5.2 Relationship Details

Relationship settings determine how tables interact during filtering and calculations. The relationship between the calendar and stock data supports consistent date-based reporting throughout the dashboard.

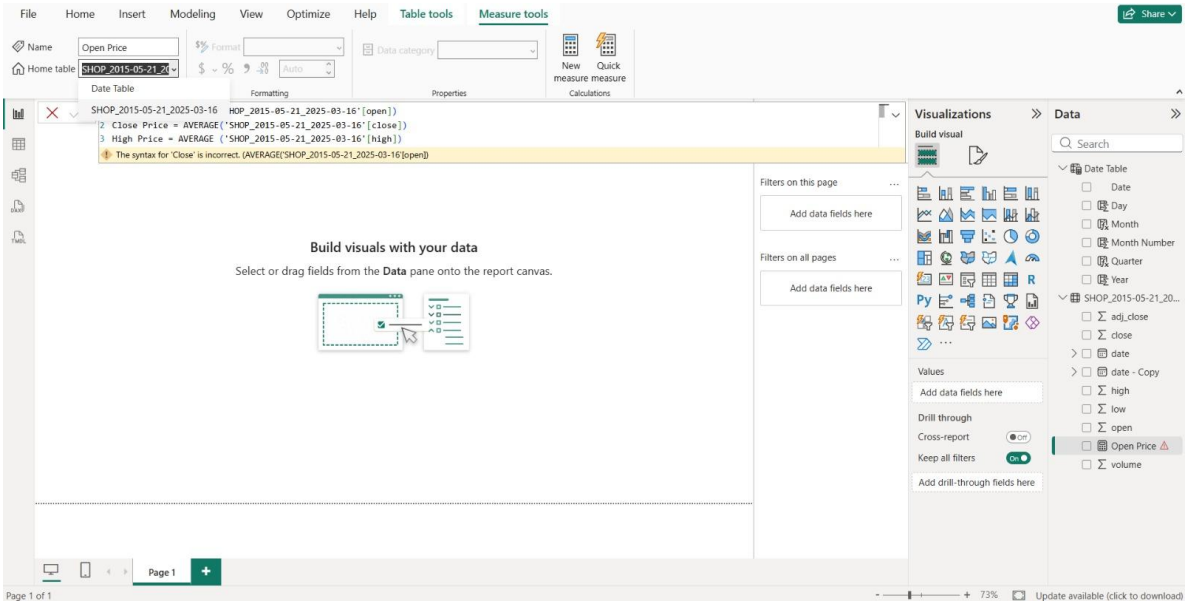


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6. Measures Creation

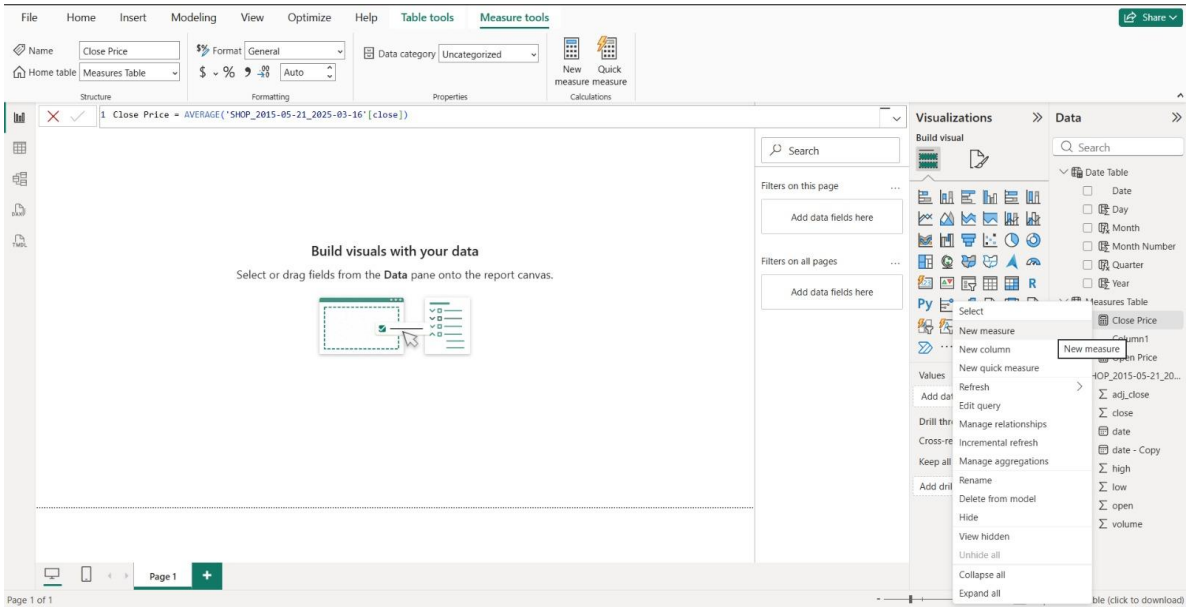
DAX measures are calculations that respond to filters and selections applied in a report. They are useful for displaying key stock values in KPI cards and summarizing data in charts.

The project includes measures associated with stock prices and trading volume. The reported indicators include Close Price, Open Price, High Price, Low Price, Close Price PY, and Trading Volume YTD.



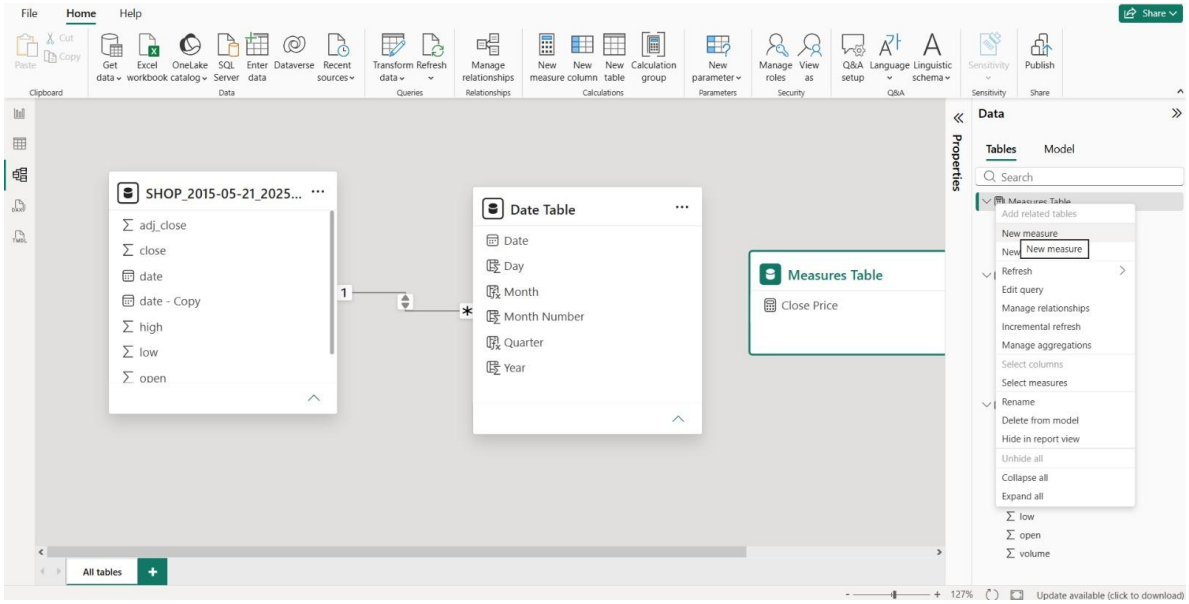
6.1 Stock Price Measures

Stock price measures summarize selected price fields for the current filter context. These measures help users compare opening, closing, highest, and lowest prices and provide concise values for KPI cards.



6.2 Additional Measures

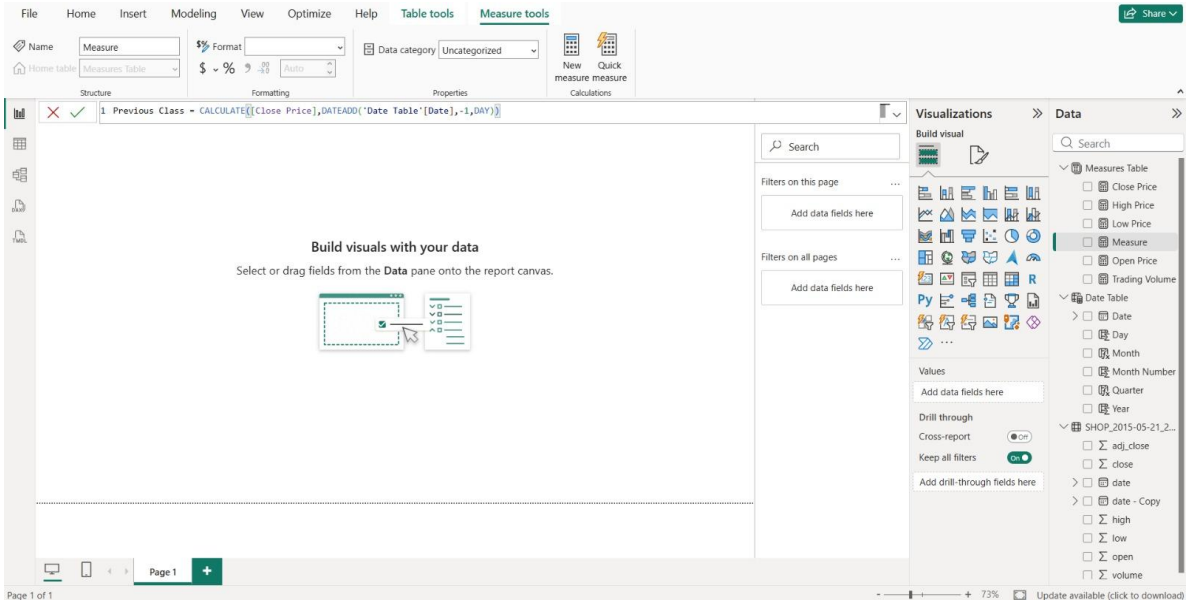
Measures can be reused across cards, charts, and report visuals. Keeping calculations as named measures makes the report easier to maintain and allows results to update when users change date filters or other selections.



7. Time-Intelligence Measures

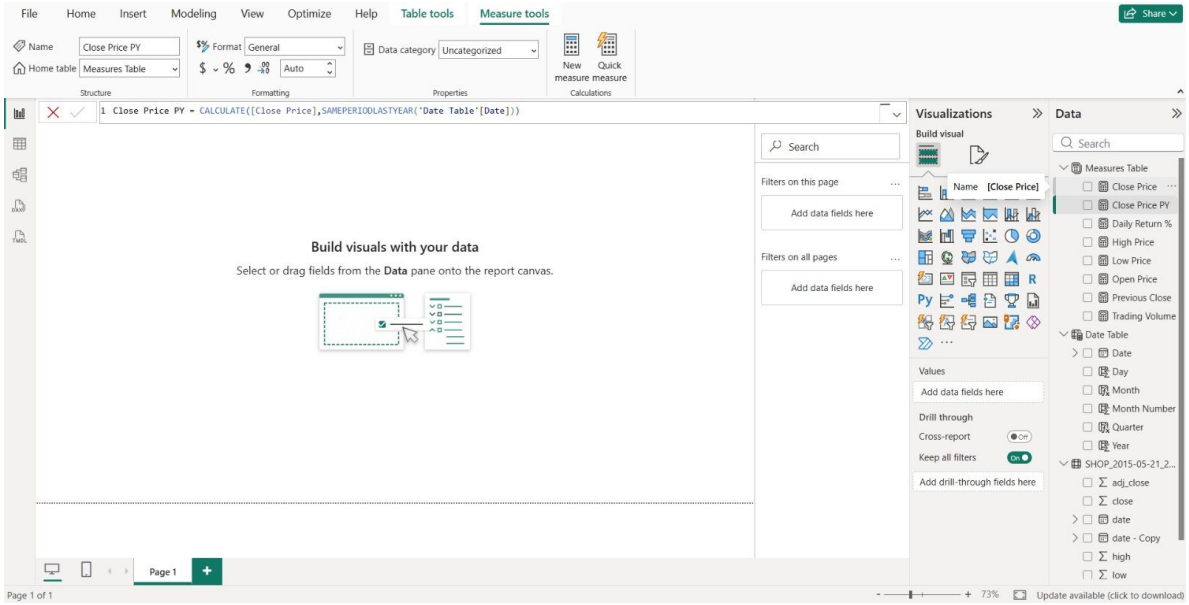
Time-intelligence measures support comparisons across periods. Close Price PY is intended to provide a previous-year comparison, while Trading Volume YTD summarizes trading volume from the beginning of the year through the selected date context.

These calculations depend on a properly configured Date Table and a working relationship between the Date Table and the stock dataset.



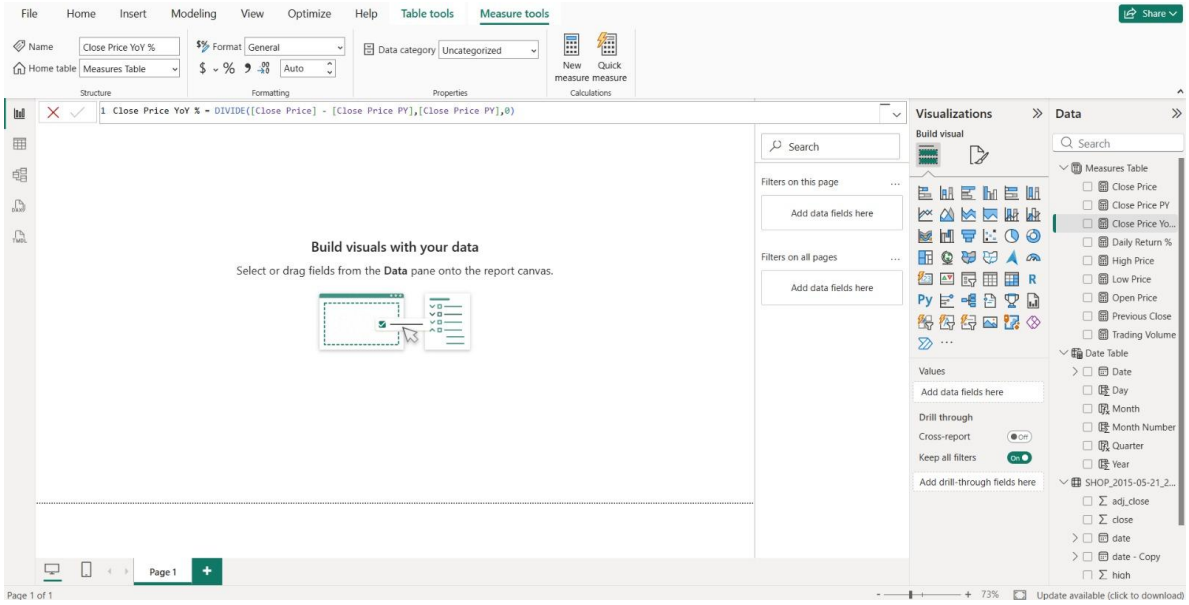
7.1 Previous-Year Comparison

Previous-year calculations compare a selected period with the corresponding period in the prior year. This can help users identify changes in stock-price performance over time, provided the date filters and measure definitions are configured correctly.



7.2 Year-to-Date Trading Volume

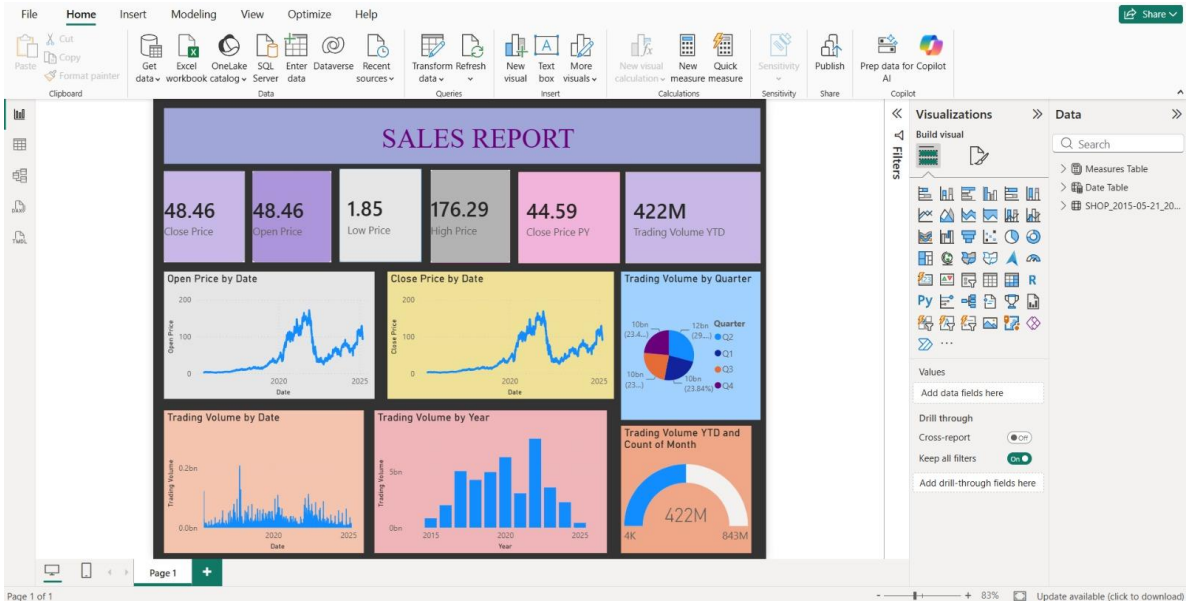
Year-to-date analysis accumulates trading volume from the start of the calendar year through the selected date. This provides a cumulative view of trading activity and can be compared across periods when the date context is set appropriately.



8. Dashboard Development

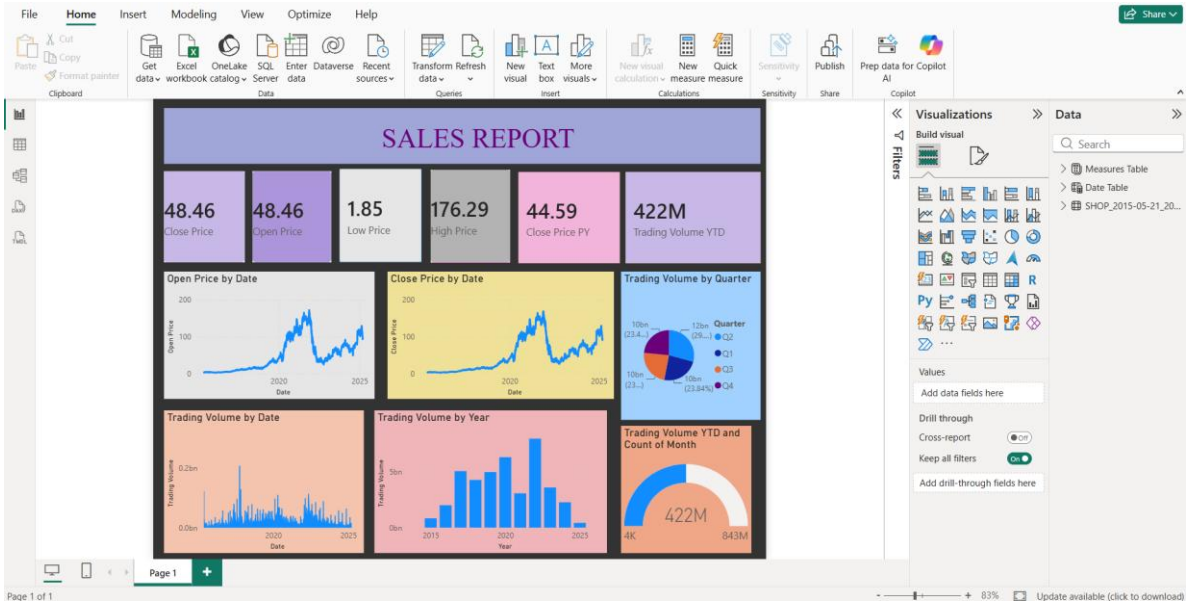
The dashboard brings together KPI cards and visualizations to present the main information in one place. KPI cards highlight important values, while line and column charts show changes and comparisons over time. Pie and gauge visuals provide additional ways to present selected aspects of the data.

A useful dashboard should have clear titles, readable labels, consistent formatting, and visuals that match the question being analysed. Date fields and filters allow users to explore the data interactively.



8.1 Final Dashboard Layout

The final dashboard combines key indicators and charts so users can review stock information quickly. The visuals support an overview of stock prices and trading activity, while date fields and measures provide the basis for comparing performance over time.



9. Dashboard Analysis

The dashboard includes KPI cards for Close Price, Open Price, High Price, Low Price, Close Price PY, and Trading Volume YTD. Visuals include line charts, column charts, a pie chart, and a gauge chart to analyse stock performance and trading activity.

The KPI cards provide a quick summary of the selected data, while the charts help reveal changes and comparisons across the reporting period. Previous-year and year-to-date measures add a time-based perspective to the analysis.

10. Learning Outcomes

This project practised the main Power BI workflow: importing and preparing a dataset, checking data types, creating a Date Table, adding calculated date columns, establishing relationships, creating DAX measures, applying time-intelligence calculations, and arranging visuals into a dashboard.

The project also demonstrates why data modelling and date preparation are important. A reliable model helps measures respond correctly to filters and supports clearer reporting.

11. Conclusion

The Week 3 project developed a Stock Market Sales Report Dashboard using Power BI and the Shopify historical stock dataset. The work covered data preparation, calendar modelling, relationships, DAX measures, time-intelligence calculations, and dashboard visualization.

The completed dashboard provides a visual summary of stock prices and trading activity and creates a foundation for further analysis of historical market patterns.